08-27 第2篇

阅读理解（说明文） Why Do Leaves Turn Red and Yellow in Autumn?

① In late October, when the long summer sun finally grows tired, the trees in our parks start to change clothes. The leaves of the **maple** turn bright red. The leaves of the **ginkgo** turn a warm, shining yellow. Even the leaves of the old **poplar** on our street become golden for about two weeks before they gently fall. Autumn is so beautiful that many poets have written about it, and many tourists travel far just to see a forest in these few weeks. But **why do leaves change colour at all**? The answer is a surprisingly clever story of sunlight, temperature and a very special chemical.

② To understand the change, we must first understand the summer green. In spring and summer, each leaf is full of a green **chemical** called **chlorophyll**, which lets the leaf use sunlight to make food for the tree. There are actually other colours inside the leaf all year round — **yellow**, **orange**, and sometimes even **brown** — but they are hidden. The strong green of chlorophyll is so **dominant** that we see only green, just as a bright lamp in a small room hides the softer light of candles.

③ In autumn, two things begin to change. First, the **days grow shorter**. Each day, the leaf receives a little less sunlight than the day before. Second, the **temperature drops**, especially at night. Most trees are very sensitive to these two signals. Their bodies quietly say, "Winter is near. We must save our energy."

④ How does a tree save energy? It **stops** making new chlorophyll in its leaves. The old chlorophyll that is still inside slowly **breaks down**. This is the key point: when the green chemical fades, the hidden yellows and oranges that were always there finally **show themselves**. The leaf has not become yellow overnight; it has simply taken off its heavy green coat so that its quieter colours can be seen. That is why a ginkgo leaf turns gold so quickly once the cold nights begin.

⑤ What about the bright red of the maple leaf? Red is slightly different. The red comes from **a new chemical** that the tree actually produces during autumn, called **anthocyanin**. Scientists are still not sure why some trees make so much of it. One popular idea is that the red pigment helps to **protect the leaf** from strong autumn sunlight during the final busy days, while the tree pulls last bits of food out of the leaf. Whatever the reason, a sunny, cool autumn usually produces more red leaves than a cloudy, mild one.

⑥ Finally, the leaf has one last job to do. Before it falls, a very thin, **corky layer** slowly grows between the leaf and the branch. This layer is like a small door that closes the leaf off from the tree. Once the door is closed, the tree no longer sends water or food to the

leaf. A gentle wind or a cool rain is then all that is needed to bring the leaf softly down to the ground.

⑦ On the ground, the story does not end. Fallen leaves slowly **rot** and become part of the soil. They return their minerals to the tree's roots and to the other small plants around. In this quiet way, each autumn is also a preparation for next spring.

⑧ Of course, not all trees change colour. **Pine** trees and some other **evergreens** keep their green needles, because their leaves are small, thick, and protected against cold air. But for many trees in the north, autumn is their short yearly show.

⑨ So next time you walk under a golden ginkgo on a cool October afternoon, remember that the leaf is not "dying beautifully". It is doing something very wise: saving energy, pulling back food, and putting on, for just two or three weeks, the colours it has been quietly carrying under its green coat all year.

注释： maple 枫树；ginkgo 银杏；poplar 杨树；chlorophyll 叶绿素；anthocyanin 花青素；corky layer 栓质层；evergreen 常绿植物

根据文章内容，选择最佳答案。

1. Why do leaves look green in summer?

A. Because green is painted on the leaves by rain.
B. Because the green chemical chlorophyll is so dominant that it hides the other colours inside the leaf.
C. Because leaves have no other colours inside them.
D. Because summer sunlight turns everything green.

2. What two signals in autumn tell the tree to "save energy"?

A. Heavy rain and long storms.
B. Days growing shorter and temperatures dropping, especially at night.
C. New flowers blooming and birds singing.
D. Students going back to school.

3. Why do ginkgo leaves turn yellow?

A. Because the tree quickly paints them yellow.
B. Because the green chlorophyll breaks down in autumn, and the yellow colour that was hidden all year finally shows itself.
C. Because the leaves drink extra yellow water.
D. Because young insects live inside each leaf.

4. Where does the bright red colour of maple leaves come from?

A. From yellow chlorophyll mixing with brown.
B. From a new chemical called **anthocyanin** that the tree actually produces during autumn.
C. From the dust on the leaves in late autumn.
D. From sunlight burning the leaves red.

5. What is the writer's main attitude toward the colour change of leaves?

A. Sad — the writer sees autumn as a painful ending.
B. Admiring — the writer sees the change as a wise and clever process of saving energy and quiet beauty.
C. Worried — the writer thinks the colour change is a sign of a sick tree.
D. Indifferent — the writer does not really care about trees.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "chlorophyll ... so dominant that we see only green"
【解析】叶子内其实还有其他颜色，但被绿色掩盖。A、C、D 均不符。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "the days grow shorter ... the temperature drops"
【解析】原文明确两信号。A、C、D 均无依据。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④段 "when the green chemical fades, the hidden yellows and oranges ... finally show themselves"
【解析】叶绿素褪去后，本来就存在的黄色显露。A、C、D 均与原文不符。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑤段 "a new chemical ... called anthocyanin"
【解析】红色来自秋天新合成的花青素。A、C、D 均与原文不符。故选 B。

5.
【答案】B
【题型】写作态度题

【定位】第⑨段 "doing something very wise: saving energy, pulling back food, and putting on ... the colours it has been quietly carrying"
【解析】作者语气中充满敬意与赞叹。A、C、D 均与作者态度不符。故选 B。